

AI-Assisted Development Workflow

A dual-track approach: instructional task development first, then app construction.



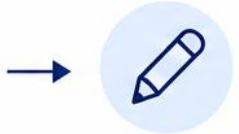
1. AI-Assisted Educational Task Development (Primary Focus)



1. Define Task Model
e.g., quick, discrete PR feedback



2. Select Clinical Case
Use a previously-vetted case with a classic presentation



3. Write & Refine Prompts
Draft prompts and refine with AI for clarity and alignment



4. Use Prompt Chaining
Break large tasks into smaller steps; link outputs to focus the model



5. Iterative Testing Across Response Types

- Highly variable responses
- Expected responses
- Remote / off-target responses

Iterate until output is consistent and effective



6. Reliable Educational Outputs
Consistently produce the intended, high-quality output

Outputs Supporting Learning



Conversational Agents



Automated Feedback



Scaffolding for Future PRs



Reflection Prompts



2. AI-Facilitated App Construction



1. Specify Inputs

- Task model (detailed)
- Programming language: R
- Framework: Shiny
- Features & functionality



2. Generate App Code

AI generates the initial Shiny application code.



3. Debug & Refine

Use AI-supported debugging and iterative refinement as needed.



4. Deploy Application

Application successfully deployed and ready for use.



Guiding Principles



Iterative & Cyclical
Continuous refinement at every stage



Evidence-Informed
Grounded in literature, pedagogy, and SME & student input



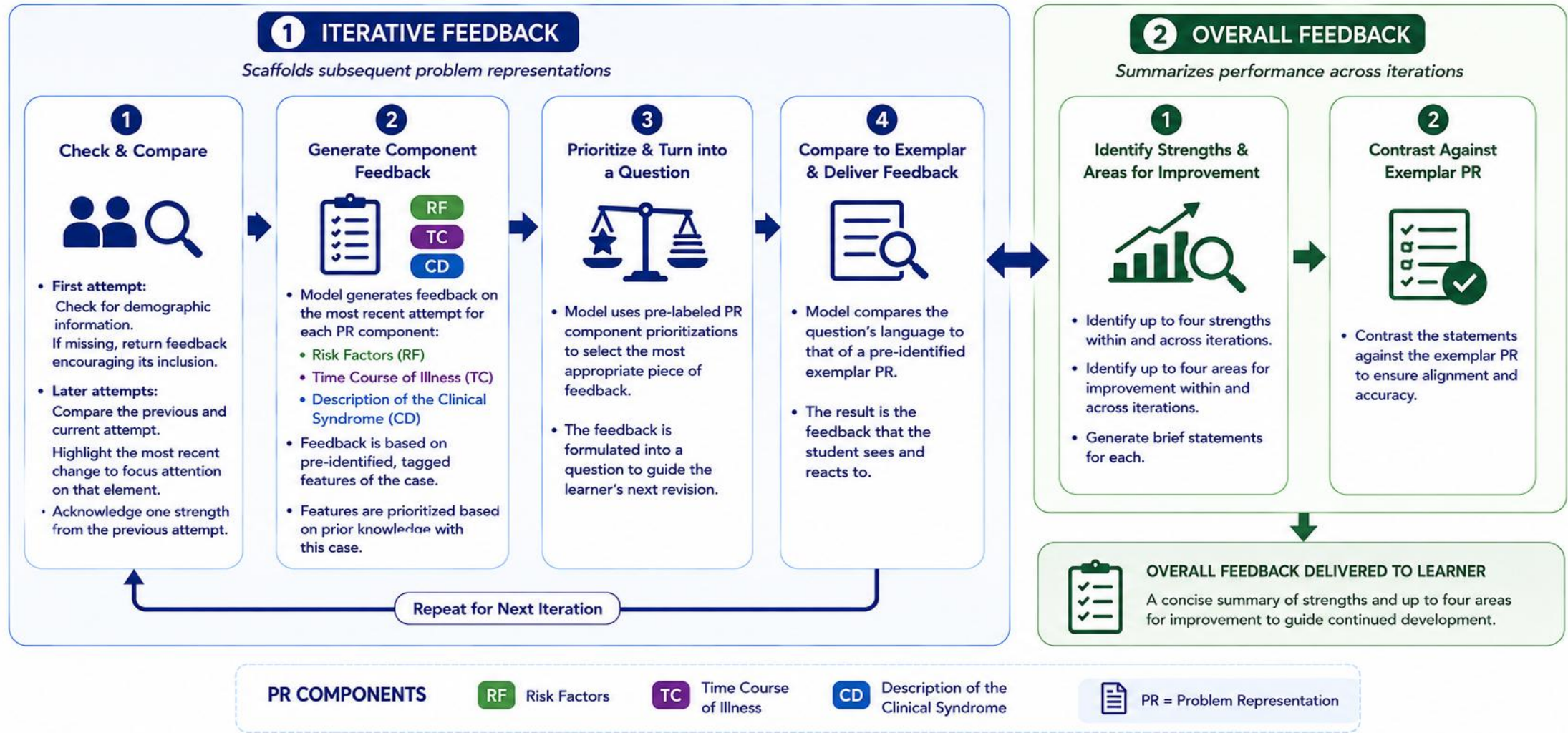
Quality & Reliability
Tested across diverse responses to ensure consistent performance



Human-in-the-Loop
Expert oversight guides prompting and development decisions

AI-ASSISTED DEVELOPMENT PROCESS: WITHIN-APP TASKS

Two Within-App Tasks



PR COMPONENTS

RF Risk Factors

TC Time Course of Illness

CD Description of the Clinical Syndrome

PR = Problem Representation